My Data My Story

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Submitted for the Mini Project #3 of CAP776

Context Setting

Intuition

Analyzing daily physical activity data can reveal meaningful insights into a person's lifestyle, fitness level, and health trends.

Objective

The goal is to understand the variation in daily step counts, detect trends or patterns, and provide insights for potential health and fitness improvements.

Challenges

Missing Data, Outliers, External Factors.

The data captured

Data description

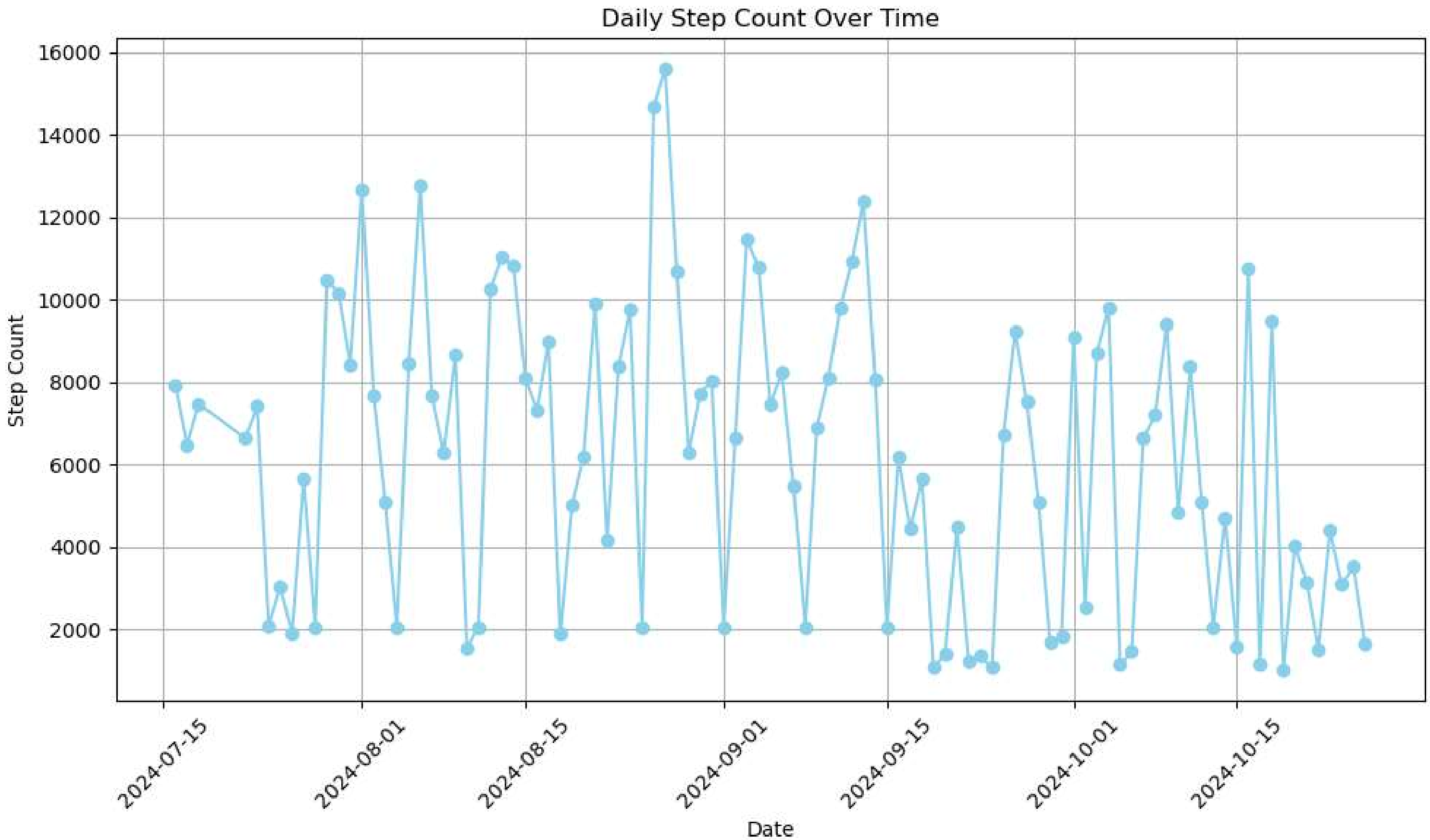
Describe your data using df.head() and df.info()

	Date	Calories (kcal)	Distance (m)	Step count
0	2024-07-16	1454.500000	4962.084056	7926.0
1	2024-07-17	1543.292471	10166.573300	6482.0
2	2024-07-18	1558.651338	3821.736487	7473.0
3	2024-07-22	1759.385401	NaN	NaN
4	2024-07-23	1659.249949	1054.445278	7409.0

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 100 entries, 0 to 99
Data columns (total 4 columns):
 #   Column           Non-Null Count  Dtype
---  ------           --------------  -----
 0   Date             100 non-null    datetime64[ns]
 1   Calories (kcal)  100 non-null    float64
 2   Distance (m)     84 non-null     float64
 3   Step count       97 non-null     float64
dtypes: datetime64[ns](1), float64(3)
memory usage: 3.3 KB
```

Story board #1

Graphical representation of data



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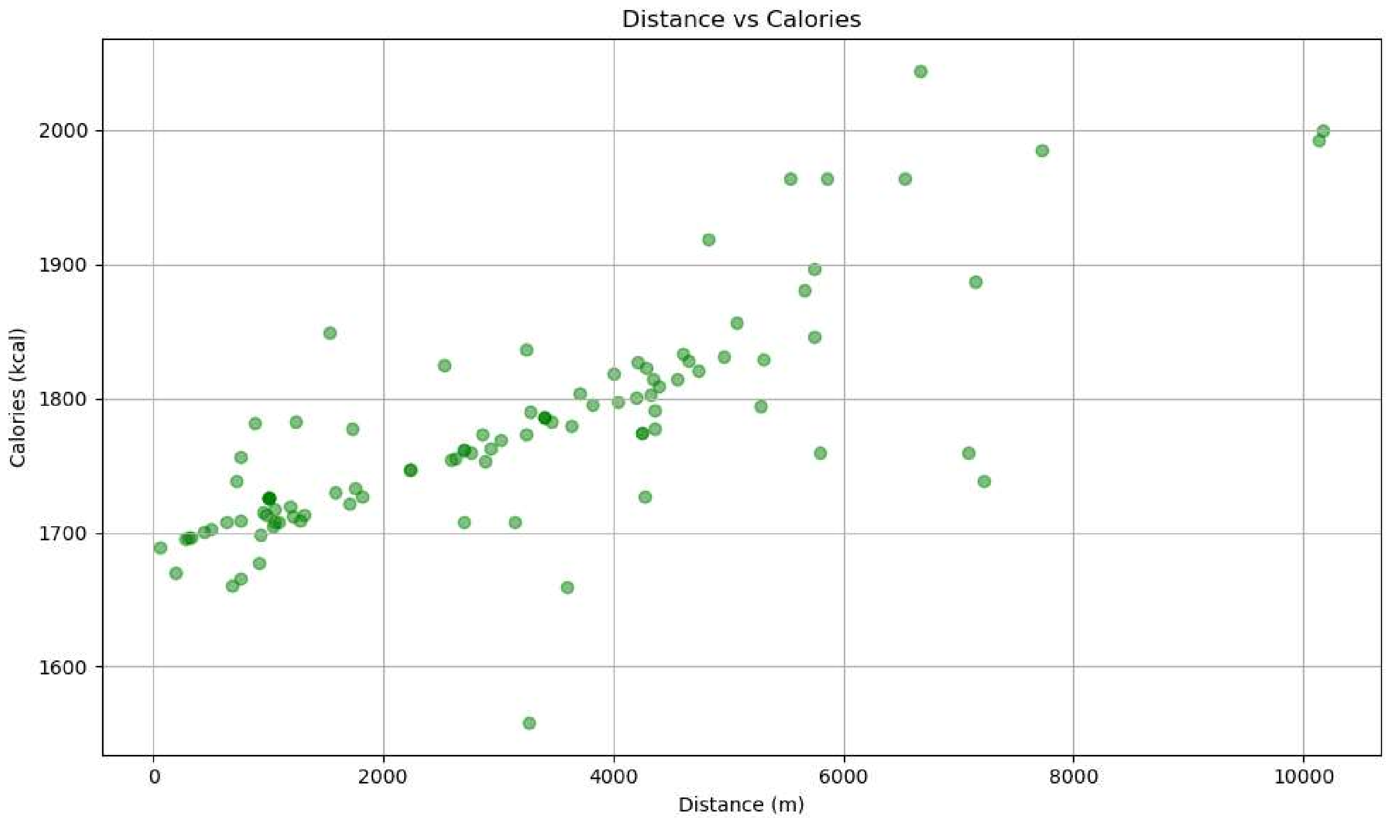
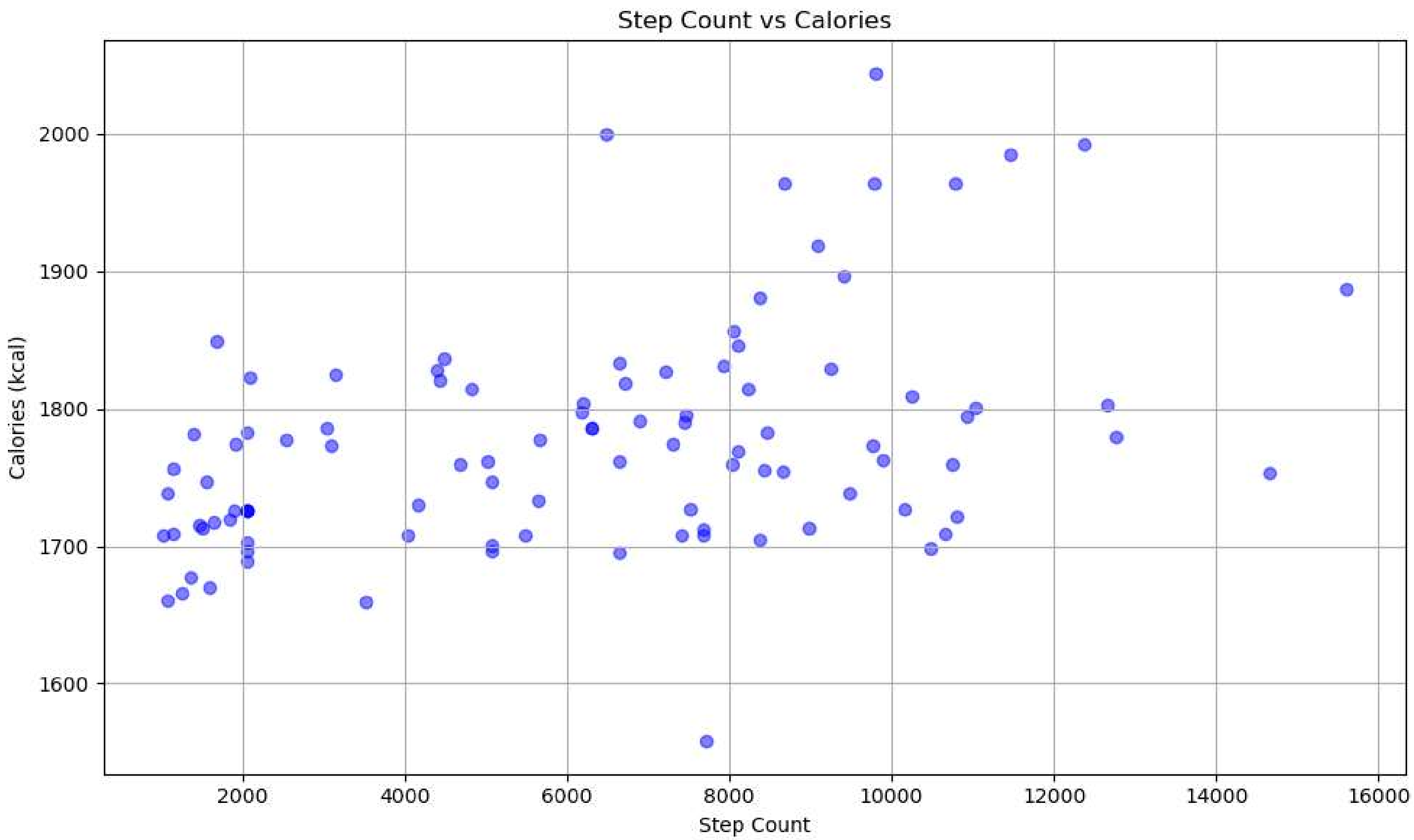
Story board #2

Graphical representation of data



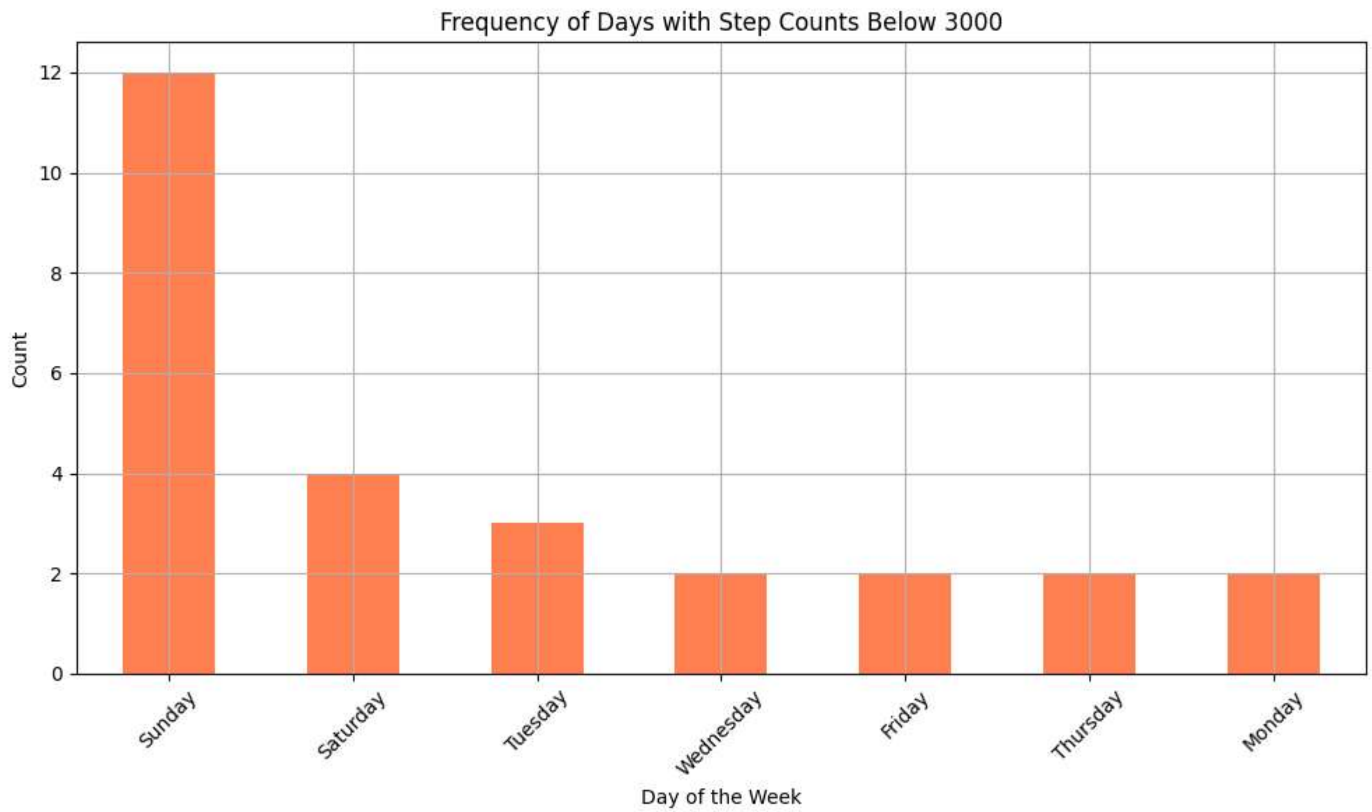
Story board #3

Graphical representation of data



Story board #4

Graphical representaton of data



Inferences drawn

Step Count Goal*: The average step count (~6,000 steps) falls below the often recommended daily target of 10,000 steps. There's room for improvement if increasing daily activity is a goal.*

Weekly Patterns: Binning by weeks shows if there's any consistent rise or drop in activity. If weekends or specific weekdays consistently have lower step counts, that could indicate lifestyle or schedule-related inactivity.

Outliers and Rest Days: Days with step counts as low as 1000 steps likely represent days of inactivity or potential errors in data collection. These should be excluded from some forms of analysis to prevent skewed results.

Future scope [Optional]

- **Personalized Recommendations**: Generate customized fitness or dietary recommendations aligned with activity patterns to support health and wellness goals.
- **Time-Series Analysis**: Perform time-series analysis to identify activity trends over time and assist in setting personal fitness goals.
- **Predictive Modeling**: Develop predictive models to forecast calories burned and activity levels based on historical data.
- **Activity Classification**: Classify days as either sedentary or active based on metrics like step count, heart rate, and speed, enabling better activity tracking.

Thank You!